

Cartesian Types for CIS I

Nominal Data Types

```
p = Point(x,y,z)
R = Rot(axis,angle)
F = Frame(R,p); R = F.R; p=F.p
```

```
p3 = p1 + p2
p2 = R*p1
p2 = F*p1 = F.R*p1+F.p
```

```
R3 = R1*R2
```

Uncertain Data Types

```
uv = uVector(v,covDeltav); %% V can be any dimension
up = uPoint(p,covEpsilon)
uR = uRot(R,covAlpha)
uF = uFrame(F,covEta)
```

```
uV = uVector(v);           %% Assumes covDelta = 0
up = uPoint(p);           %% Assumes that covEpsilon = 0
uR = uRot(R);             %% Assumes that covAlpha = 0
uF = uFrame(F);           %% Assumes that covEta = 0
```

```
up3 = up1+up2
up3 = p1+up2
up2 = uR*up1
up2 = uR*p1
```

```
uR = uRot(axis,uAngle)    %% uAngle is an uncertain 1 vector
uR = uRot(uAxis,Angle)    %% uAxis is an uncertain 3 vector or an uncertain point
uR = uRot(uAxis,uAngle)   %% uAxis is an uncertain 3 vector or an uncertain point
```

```
uR3 = uR1*uR2
uR3 = R1*uR2
uR3 = uR1*R2
```

```
uF = uFrame(uR,up)
uF = uFrame(R,up)
uF = uFrame(uR,p)
```

$$uF3 = uF1 * uF2$$

$$uF3 = R1 * uF2$$

$$uF3 = uF1 * R2$$